Multi-Horizon Joint-Embedding Predictive Architectures and Coordinate Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Reconstructive time-series anomaly detectors score raw observations, making them sensitive to benign sensor jitter and drift. Joint-Embedding Predictive Architectures (JEPAs) forecast normalized latent window summaries, yet collapsing coordinates into one scalar residual allows drifting coordinates to inflate thresholds. We introduce two multi-horizon circuit-breaker models: Laminar-JEPA, a four-horizon, 224-dial panel combining sub-horizon magnitudes, drift, fluctuation, endpoint differences, and 32 onset relays with optional nominal weight-space quenching; and Laminar-JEPA-Lite, a distilled three-horizon, 128-dial position-plus-drift detector using a parallel single-normalization predictor and channel-gated spectral splitting with zero weight edits. Laminar-JEPA-Lite cuts parameters by 35 to 52 percent and accelerates inference by 2.43 times. On the 33-stream development benchmark, Laminar-JEPA and Laminar-JEPA-Lite hold autonomous causal false-positive rate (FPR) at 1.3 and 1.0 percent versus 22.2 to 24.3 percent for reconstructors, reaching buffered point-wise F1 of 0.244 and 0.242 and seven-dataset macro F1 of 0.133 and 0.171. On the 19-stream residual cache, the 224-dial panel lifts causal F1 from 0.013 to 0.071 over scalar collapse, and a 160-dial subset reaches 0.098. Across 118 untouched held-out streams under disjoint checkpoint selection, Laminar-JEPA-Lite lifts causal F1 from 0.074 to 0.085 and event recall from 49.5 to 67.0 percent over scalar multi-horizon scoring while remaining positive across all 12 leave-one-dataset-out folds on 141 streams, whereas quenching lowers four-horizon causal FPR across seven of nine datasets. Calibration precision depends on autocorrelation-adjusted sample size; causal ramp detection vanishes at four context windows, and single-sensor sensitivity raises false alarms.**

***Index Terms*—Anomaly detection, Multivariate time series, Representation learning, Self-supervised learning, Statistical process control.**

I. Introduction

Spacecraft, chemical plants, server farms, and wearable sensors report health across dozens of channels [1]–[4]. Because labeled faults are scarce, a streaming monitor must learn nominal behavior from healthy history alone, flag anomalies promptly, and keep false alarms low during normal operation.

Why Observation-Space Reconstructors Suffer High False-Alarm Rates: Reconstructors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] score raw sensor error. Because raw telemetry mixes jitter, thermal drift, and disparate channel scales, summing errors across 25–55 channels produces heavy-tailed spikes. Calibrated on healthy windows via Streaming Peaks-Over-Threshold (SPOT) [8], reconstructors fire during 22%–31% of later healthy data; a rate-matched coin flip catches 60.0% of anomaly events, exceeding TimesNet's 58.8%.

The Bottleneck of Scalar Score Aggregation: Joint-Embedding Predictive Architectures (JEPAs) [9], [10] compress each window into a 32-dimensional normalized summary and forecast future summaries in latent space, cutting nominal false alarms to 0.6%–2.4%. Yet scalar latent detectors still sum all coordinate errors against one threshold: if one or two coordinates shift slightly between training and calibration (*hijacker dials*), they inflate the global threshold and mask faults on the other 30 coordinates (causal Point-F1 0.050–0.055, delay 138–154 steps).

*Multi-Horizon Coordinate Circuit Breakers (*Laminar-JEPA *and* Laminar-JEPA-Lite*):* We pair multi-horizon JEPA forecasting with calibrated Boolean circuit-breaker panels in two tiers (Fig. 1): (1) *Laminar-JEPA* maps context and target windows into 32D layer-normalization (LayerNorm) [11] summaries with resolvent residual whitening, expands each coordinate's four-horizon error trajectory into $7 \times 32 = 224$ dials plus 32 right-edge onset relays, and applies optional nominal split-gated weight-space quenching; and (2) *Laminar-JEPA-Lite*, distilled from a layer-by-layer activation and component autopsy, replaces sequential four-horizon rollout and post-hoc weight editing with a parallel three-horizon single-LayerNorm predictor, channel-count-gated spectral splitting ($K \geq 25$), and $4 \times 32 = 128$ position-plus-drift breakers under a single nominal quorum, cutting parameters by 35%–52% and accelerating windowed inference by $2.43\times$.

A. Contributions

We separate evaluation into a 33-stream development cohort (plus a 19-stream residual cache) and a disjoint 118-stream untouched held-out cohort within a 141-stream archive across 12 datasets under disjoint checkpoint selection (Section V-B):

1) *Flagship and distilled panels:* On the 33-stream benchmark, *Laminar-JEPA* and *Laminar-JEPA-Lite* reach buffered Point-F1 of 0.244 and 0.242 and causal event recall (ER) of 67.3% and 65.0% at 1.3% and 1.0% held-out false-positive rate (FPR), with *Laminar-JEPA-Lite* leading 7-dataset macro Point-F1 (0.171 vs. 0.133). On the 19-stream

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

Color versions of one or more figures in this article are available online at https://ieeexplore.ieee.org.

cache, the 224-dial panel lifts causal Point-F1 from 0.013 to 0.071 over 1D collapse ($p = 0.0023$), and a 160-dial subset reaches 0.098. Across 118 untouched streams ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$), *Laminar-JEPA-Lite* raises causal Point-F1 from 0.074 to 0.085 ($p = 0.0034$; $+0.025$ over TS-JEPA, $p = 1.6 \times 10^{-4}$), lifts causal ER from 49.5% to 67.0% across $8/9$ datasets (sign-flip $p = 0.012$), and stays positive in all $12/12$ leave-one-dataset-out (LODO) folds on $n = 141$, while nominal quenching lowers 4H causal FPR across $7/9$ datasets ($p = 0.016$).

2) *Autopsy distillation, mechanisms, and latency:* Matched-backbone controls (Table I(c)) isolate LayerNorm tail compression and barycenter drift reduction while showing scalar spectral reweighting is null. Layer activation traces, LayerNorm spillover audits, and a $K \geq 25$ spectral threshold motivate *Laminar-JEPA-Lite*. Across $S \in \{1, 4, 16, 32, 64\}$ on five exploratory streams, both panels reach 2.1–4.3$\times$ the scalar causal Point-F1 at $S = 1, 4$ without establishing cross-domain latency rules.
3) *False-alarm control and contextual faults:* On the 19 non-GHL streams, latent prediction holds autonomous causal FPR at 0.6%–2.4% versus 22.2%–24.3% for reconstructors ($N = 7$ dataset $p = 0.016$). At diagnostic budget $f^* = 1\%$ test FPR, *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% and 56.0% of events versus 21.7% for TimesNet, and reach 0.519–0.532 buffered precision-recall (PR) AUC with 77.5%–87.5% autonomous ER on NASA contextual faults (Tables III–IV, Fig. 4).
4) *Calibration sample size, single-channel branch, and limits:* Proposition 1 bounds horizon-fluctuation concentration $R_{\text{LT}} \in [\frac{1}{M-1}, 1]$, while scalar R_{LT} reweighting is null (ΔF1 $= -0.005$). Sokal autocorrelation (τ_{int} up to 206) and block-bootstrap audits show $N_{\text{eff}} = 300$ does not guarantee 99.9th-percentile precision; interquartile-range (IQR) and Beta-smoothed quantiles remain exploratory. Across 19 streams, an optional channel branch lifts injected single-sensor ER to 80.7%–96.5% while raising nominal causal FPR from 1.5% to 6.8%–19.4%; natural sensor failures and slow ramps ($L \geq 4C$) remain limitations.

Roadmap. Section II reviews related work, Section III details the benchmark, Section IV formulates *Laminar-JEPA* and *Laminar-JEPA-Lite*, Section V defines the protocol, Sections VI–VII report results, and Section VIII concludes.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep detectors typically forecast or reconstruct raw windows and threshold $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$: autoregressive long short-term memory (LSTM) forecasters [4] model spacecraft channels directly; TranAD [6] pairs adversarial training with self-conditioning attention; Anomaly Transformer [7] contrasts Gaussian prior and series-level attention; and TimesNet [5] reshapes 1D series along Fast Fourier Transform (FFT) periods for 2D convolution. Equal channel-wise reconstruction losses fit stochastic jitter, so summing errors across $K \geq 25$ channels yields heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) spikes and high false alarms under baseline shift [12].

Contrastive and Joint-Embedding Predictive Architectures: Representation methods encode windows into compact latents [13]. Contrastive approaches—temporal-contextual contrasting [14], TS2Vec [15], Neural Contextual Anomaly Detection (NCAD) [10] with synthetic anomalies, and dual-attention DCdetector [16]—depend on whether synthetic perturbations or patch discrepancies match test faults (0.088 and 0.097 stream Point-F1 for NCAD and DCdetector vs. 0.186 for latent prediction on 130 streams, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [17], [18] predict target embeddings z_{tgt} from context z_{ctx} without observation decoding or negative pairs, avoiding collapse via exponential moving average (EMA) targets [19] or variance-covariance terms [20]–[22].

Residual Decomposition, Conformal Thresholding, and Statistical Process Control: Telemetry reflects underlying system dynamics [23]. Neural ordinary differential equations (ODEs) [24], Physics-Informed Neural Networks (PINNs) [25], Hamiltonian/Lagrangian nets [26], [27], and Koopman lifts [28]–[32] model structured dynamics, though partial differential equation (PDE) penalties require known laws and exhibit stiff gradients on uncalibrated sensors [33]; we instead borrow the algebraic mean-plus-fluctuation identity of Reynolds averaging [34] strictly as a multi-horizon residual summary. Multivariate statistical process control (SPC) charts such as Hotelling's T^2 [35] and Multivariate Exponentially Weighted Moving Average (MEWMA) [36] collapse whitened coordinates into one scalar, as do 1D deep scores calibrated via extreme-value theory [37] and Generalized Pareto Distribution (GPD) tails in SPOT [8] under serial dependence [38]. Conversely, univariate Cumulative Sum (CUSUM) banks [39] and per-coordinate split-conformal rules under Bonferroni, Šidák, or Benjamini–Hochberg (BH) false-discovery-rate (FDR) control [40] hit a discrete calibration floor $(N_{\text{cal}} + 1)^{-1}$ on short histories, whereas matched-rate per-coordinate and recalibrated Mahalanobis conformal rules inflate under coordinate drift (Section VI-A). All methods are evaluated without point adjustment [12], [41], [42].

III. Multi-Sensor Telemetry Benchmark Suite

Our benchmark suite spans 45 continuous telemetry streams across 11 datasets from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [12], and mTSBench [45], expanded in archive-wide evaluations to 141 qualifying multivariate streams across 12 datasets (Section V-B). Thirty-three streams enter the development protocol (19 non-GHL streams and 14 GHL streams) and 12 are reserved with explicit accounting below. *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [46] (K=38), where NASA [4] labels each SMAP/MSL sequence as an out-of-range *point* fault or in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [43] at 64 Hz (K=9, 72 episodes, where S=64 equals 1 s). *Process and infrastructure telemetry* covers 14 Gasoil Heating Loop

(GHL) streams (K=16, 26 slow-ramp faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry (K=9, 37 anomalies), Metro I-94 traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72). Section VI-B evaluates six injected single-channel fault classes across all 19 multivariate development streams ($K \in [5, 55]$).

Each stream provides nominal history ($N_{\text{tr}} \in [2{,}074, 120{,}000]$) and labeled test data ($N_{\text{test}} \in [2{,}158, 114{,}001]$), standardized per channel on the weight-fitting prefix ($\boldsymbol{\sigma}$ floored at 10^{-2}) and split chronologically into training (65%, further split $50\%/15\%$ into weight-fitting and checkpoint-selection prefixes under disjoint selection; Section V-A), nominal calibration (17.5%), a 64-step isolation gap, and held-out nominal evaluation (17.5%). Development 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$), $K \leq 55$, and $N_{\text{test}} \leq 1.2{\times}10^5$ (Table III(c)); the 12 reserved streams comprise five short-nominal streams ($T_{\text{nom}} \in [764, 1{,}492]$, $N_{\text{cal}} \leq 101$: CalIt2, MSL C-2, MSL D-16, two Room-Occupancy), six CICIDS streams (K=72), and GECCO (K=9, included in archive cohorts). Archive cohorts evaluate all 141 qualifying streams across 12 datasets ($N_{\text{cal}} \geq 100$, ≥ 1 anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN; 118 untouched by design).

Two caveats bound external validity: SMAP and MSL contain documented labeling flaws [42], and streams within multi-stream archives share hardware platforms. Dataset clusters ($N \in \{7, 9, 12\}$) are therefore treated as co-primary units alongside streams via Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and LODO audits (Section V-A).

IV. Methodology: Multi-Horizon Laminar-JEPA and Distilled Laminar-JEPA-Lite

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T\times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and splits the stream at index t into a historical *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C\times K}$ ($C = 256$) and a future *target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S\times K}$ ($S = 64$, up to causal decision timestamp $\tau = t + S - 1$). To separate slow drift from high-frequency vibration before cross-channel mixing, multichannel inputs ($K > 1$) pass through a causal depthwise 1D convolutional stem ($K \leq 1$ uses an identity bypass) and are split across $J = 4$ smooth dyadic shells on a 256-point real fast Fourier transform (rFFT) grid normalized into an amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), filtered by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [48].

Unlike reconstructive detectors that penalize raw waveform error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$, we map windows into a compact 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ to approximately radius-$\sqrt{D}$ layer-normalized vectors; the normalization epsilon prevents exact sphere membership. Target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m\phi_{k-1} + (1 - m)\theta_k$ with momentum $m = 0.996$ (0.995 in archive cohorts) [17], [19]. Because deep dilated TCN stacks can smooth sharp onset spikes while non-stationary regimes shift the latent mean, Stage 1 introduces two operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* To preserve fast transient spikes alongside long-range context, a tap head extracts $L_{\text{tap}} = 3$ feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w\times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic stem, mid-depth TCN block, and terminal TCN block. Concatenating the channel-centered terminal state and temporal root-mean-square (RMS) energy of each tap forms $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, which is projected to $\mathbb{R}^D$ and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \leq 0.5$:
$$\mathbf{z}_{\text{fused}} = (1 - g)\,\mathbf{z}_{\text{deep}} + g\ \mathrm{LN}\big(W_2\,\phi_{\text{GELU}}(\mathrm{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$
2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ (updated at rate 0.05 on training batches) is copied synchronously to the EMA target encoder and subtracted from both context and target vectors prior to terminal LayerNorm, $z = \mathrm{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}})$. This attenuates learned common-mode offsets without guaranteeing invariance to unseen drift.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$, trained with predictive mean-squared error (MSE) and a log-determinant volume term $-\frac{\beta}{D}\ln\det(\boldsymbol{\Sigma}+\epsilon\mathbf{I})$ ($\beta = 0.05$) related to VICReg [20]. LayerNorm limits each vector's scale but does not bound its temporal tail kurtosis; pooling an $S = 64$ window into one vector can average out brief faults.

To resolve how a trajectory evolves across the target window, Stage 2 of *Laminar-JEPA* uses $M = 4$ nested prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ with $S_m = \min(S, \max(2, \lfloor mS/M \rfloor))$ for $m < M$ and $S_M = S$ ($\{16, 32, 48, 64\}$ at $S = 64$). Each is encoded by E_ϕ into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. At $S = 1$, all four targets contain the same observation; distinct residuals arise from the predictor rollout, not from distinct target horizons. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a recurrent sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \mathrm{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \ldots, M - 1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ predicts $\hat{z}_{\text{tgt}}^{(M)} = \mathrm{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon

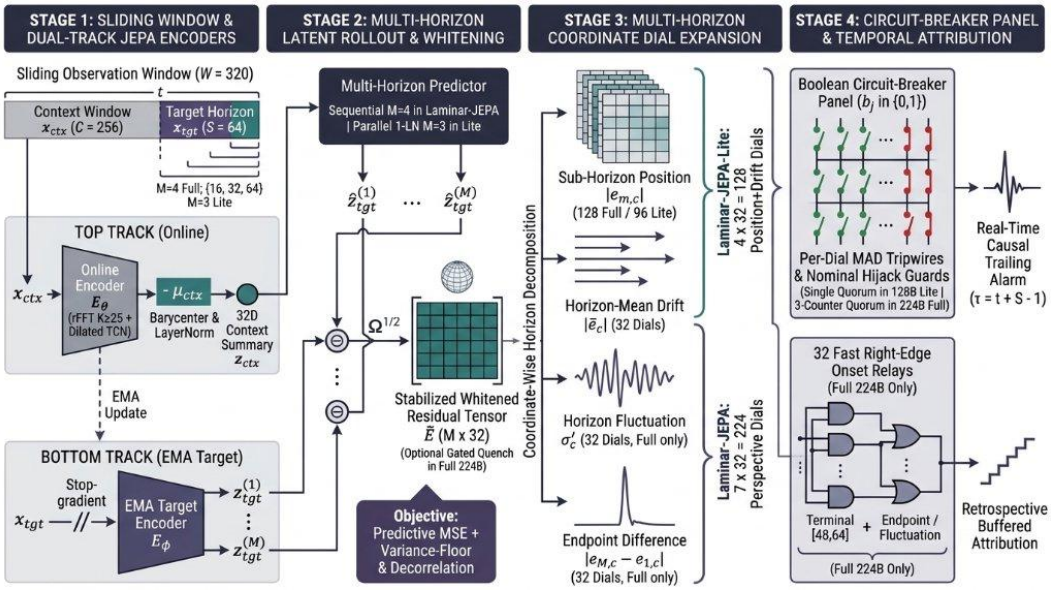


Fig. 1. Overview of the four-stage *Laminar-JEPA* and distilled *Laminar-JEPA-Lite* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with conditional rFFT splitting ($K \geq 25$ in Lite), barycenter centering, and LayerNorm; **(2)** multi-horizon rollout (M=4 sequential in full *Laminar-JEPA*; M=3 parallel single-LayerNorm in *Lite*) under variance-floor and decorrelation loss plus resolvent whitening into $\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$; **(3)** horizon expansion into $7 \times 32 = 224$ dials (full) or $4 \times 32 = 128$ position-plus-drift dials (*Lite*); and **(4)** per-dial Boolean circuit breakers emitting causal (τ=t+S−1) or prefix-buffered alarms.

MSE alongside a non-collapse matrix penalty $\mathbf{\Delta}_\gamma$:

$$\mathcal{L}_{\text{Lam}} = \sum_{m=1}^{M} \frac{w_m}{D} \left(\left\| \hat{z}_{\text{tgt}}^{(m)} - \text{sg}\left(z_{\text{tgt}}^{(m)}\right) \right\|_2^2 + \left\| \mathbf{\Delta}_\gamma\left(\hat{z}_{\text{tgt}}^{(m)}\right) \right\|_F^2 \right). \tag{2}$$

Here $\mathbf{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ penalizes collapse without requiring $B > D$. Its diagonal entries are $\sqrt{\max(0, \gamma - \sigma_d)}$, where σ_d uses the batch-biased variance and 10^{-4} stabilization ($\gamma = 0.75$). Off-diagonal entries are the sample covariances $\Sigma_{dj}/\sqrt{2}$ ($d \neq j$), computed with the $B-1$ divisor. The distinct variance conventions matter for exact correspondence with the objective. At $B = 16 < D = 32$, the measured effective rank $\text{RankMe}(z_{\text{tgt}})$ rises from 11.6 to 19.8 [49].

C. Stage 3: Calibration-Stabilized Whitening, Optional Weight-Space Quenching, and 224-Dial Expansion

Why does collapsing a multi-horizon residual tensor into a 1D scalar score degrade real-time detection? Stacking the $M = 4$ signed residuals $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ forms $\mathbf{E}(t) \in \mathbb{R}^{M\times D}$ ($d_{\text{tot}} = MD = 128$). On weight-fitting windows, the sample residual covariance $\hat{\mathbf{\Sigma}}_{\text{res}} = \mathbf{U}\,\text{diag}(\lambda_k)\mathbf{U}^\top$ defines the positive-semidefinite resolvent precision

$$\mathbf{\Omega} = \mathbf{U}\,\text{diag}\left(\frac{\lambda_k}{(\lambda_k + 10^{-3})^2}\right)\mathbf{U}^\top \in \mathcal{S}_+^{128}.$$

Applying $\mathbf{\Omega}^{1/2}$ to residuals centered by $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{128}$ gives $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M\times D}$, attenuating near-null modes without forcing unit variance on every coordinate. Nominal regime shifts between training and calibration can still inflate a few coordinates and dominate a scalar norm. Stage 3 of *Laminar-JEPA* addresses this through two mechanisms:

1) *Optional weight-space quenching and calibration-stabilized whitening:* When nominal drift between training and calibration inflates a few coordinates, *Laminar-JEPA* applies an optional zero-gradient post-training edit on training and nominal-calibration windows alone (without inverting a LayerNorm Jacobian). Each coordinate $c \in \{1, \ldots, D\}$ receives a standardized instability score r_c from its training-to-calibration mean shift δ_c and standard-deviation ratio v_c, calibration half-split shift b_c and ratio $t_c \geq 1$, nonnegative lag-ℓ autocorrelation ρ_c of the horizon-mean raw residual, and nominal peak-window trip share h_c:

$$\begin{aligned} s_c &= 0.35\delta_c + 0.30(v_c - 1)_+ + 0.45b_c \\ &\quad + 0.35(t_c - 1)_+ + 0.25\rho_c - 1.50h_c, \\ r_c &= \frac{s_c - \text{med}_j\, s_j}{1.4826\ \text{med}_j\, |s_j - \text{med}_k\, s_k| + 10^{-4}}. \end{aligned} \tag{3}$$

For candidate top-k coordinate sets $\mathcal{C}_k$ ($k \in \{4, 2\}$ of r_c), four steps execute in order: *(Q1)* damp the c-th output rows and LayerNorm gains of $\{P_{\text{sub}}, P_\theta\}$ by $a_c = \text{clip}((1 + 0.5(r_c)_+)^{-1}, 0.35, 0.85)$ and shift $\mathbf{P}_{:,c}$ by $+0.35(\boldsymbol{\mu}_{:,c}^{\text{cal}} - \boldsymbol{\mu}_{:,c}^{\text{tr}})$ (GELU and 32D LayerNorm couple coordinates, causing unselected spillover audited in Section VI-A); *(Q2)* blend $\bar{\mu}_{\text{res}}[mD + c]$ 75% toward post-edit calibration means for $c \in \mathcal{C}_k$; *(Q3)* attenuate off-diagonal precision couplings $\Omega_{m_1 D+c, m_2 D+c'}$ ($c' \neq c$) by 0.15, project onto $\mathcal{S}_+^{MD}$ (floor 10^{-6}), and rescale post-edit whitened calibration RMS toward 0.50 ($c \notin \mathcal{C}_k$) and $[0.975, 1.08]$ ($c \in \mathcal{C}_k$); and *(Q4)* retain the edit (preferring $k = 2$ on ties) iff nominal half-split FPR improves without shrinking K_{224} or K_{128} below $0.92\times$ their pre-edit values, reverting otherwise. Any coordinate with $\text{RMS}_{\text{cal}}^{\text{white}}(m, c) > 1.10$ then blends smoothly toward $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\text{RMS}_{\text{cal}}^{\text{raw}}(m, c) + 10^{-4})$:

$$\begin{aligned} \tilde{e}_{m,c}(t) &= w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t), \\ w_{m,c} &= \frac{1}{1 + 2\max\left(0, \text{RMS}_{\text{cal}}^{\text{white}}(m, c) - 1.10\right)^2}. \end{aligned} \tag{4}$$

2) *Multi-horizon expansion into $7D = 224$ dials:* Motivated algebraically by the mean-plus-fluctuation split of Reynolds averaging [34] along the discrete horizon index

$m \in \{1,\dots,M\}$ of each coordinate $c \in \{1,\dots,32\}$, we decompose $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into horizon mean $\bar{e}_c(t) = \frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and zero-mean fluctuation $F_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$, forming $7D = 224$ nonnegative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$: (a) *sub-horizon position magnitudes ($4D = 128$)* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ ($m \in \{1..4\}$); (b) *horizon-mean drift ($D = 32$)* $d_{\text{drift},c}(t) = |\bar{e}_c(t)|$; (c) *horizon fluctuation ($D = 32$)* $d_{\text{fluc},c}(t) = \sigma'_c(t) = \sqrt{\frac{1}{M}\sum_{m=1}^{M} F^2_{m,c}(t)}$; and (d) *endpoint difference ($D = 32$)* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$.

Subtracting each coordinate's horizon mean splits multi-horizon error energy orthogonally into steady drift plus zero-mean fluctuation and restricts the $M \times M$ horizon Gram matrix to $\mathbf{1}_M^{\perp}$ (describing representation geometry, not guaranteeing detection accuracy):

Proposition 1 (Horizon Energy Split and Spectral Concentration Bounds)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) denote the stabilized residual matrix with horizon-mean vector $\bar{\mathbf{e}}(t) = M^{-1}\tilde{\mathbf{E}}(t)^{\top}\mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{F}(t) = \tilde{\mathbf{E}}(t) - \mathbf{1}_M\bar{\mathbf{e}}(t)^{\top} \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise energy orthogonality and endpoint bound: *For every $c \in \{1,\dots,D\}$, $\frac{1}{M}\sum_{m=1}^{M} d^2_{pos,m,c}(t) = d^2_{drift,c}(t) + d^2_{fluc,c}(t)$ and $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Horizon Gram matrix and spectral concentration bounds: *Define the $M \times M$ horizon Gram matrix $\mathbf{G}(t) = \mathbf{F}(t)\mathbf{F}(t)^{\top} \in \mathcal{S}^M_+ \subset \mathbb{R}^{M\times M}$ (distinct from the $D \times D$ coordinate matrix $\mathbf{F}(t)^{\top}\mathbf{F}(t)$), with ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$. For any $\gamma > -1/2$ ($p = \gamma + \frac{1}{2} > 0$) and $\mathbf{F}(t) \neq \mathbf{0}$, the scale-invariant dominant-mode concentration ratio (using Lieb–Thirring power moments [50]),*

$$R_{LT}\big(\mathbf{F}(t)\big) = \frac{\lambda_M^p}{\sum_{m=1}^{M}\lambda_m^p} \in \Big[\frac{1}{M-1},\, 1\Big], \tag{5}$$

attains its lower bound $\frac{1}{M-1}$ *($= \frac{1}{3}$ at $M = 4$) if and only if fluctuation energy is isotropic across all $M-1$ nonzero horizon modes ($\lambda_2 = \cdots = \lambda_M > 0$, requiring $D \geq M-1$), and attains its* upper bound 1 *if and only if $\mathbf{F}(t)$ has rank one ($\lambda_2 = \cdots = \lambda_{M-1} = 0 < \lambda_M$). (Defining the reciprocal spread ratio $\sum_{m=1}^{M}\lambda_m^p/[(M-1)\lambda_M^p]$ swaps the two equality endpoints.) Empirically, scalar reweighting by R_{LT} has a null effect (Section VI).*

Proof. Since $\mathbf{F}$ has zero column sums ($F_{1,c} + \cdots + F_{M,c} = 0$), expanding $M^{-1}\sum_{m=1}^{M}(\bar{e}_c + F_{m,c})^2$ eliminates the cross term and gives the energy identity; applying $(F_{M,c} - F_{1,c})^2 \leq 2(F^2_{M,c} + F^2_{1,c}) \leq 2M\, d^2_{\text{fluc},c}$ proves (i).

For (ii), $\mathbf{F}^{\top}\mathbf{1}_M = \tilde{\mathbf{E}}^{\top}\mathbf{1}_M - \bar{\mathbf{e}}(\mathbf{1}_M^{\top}\mathbf{1}_M) = \mathbf{0}_D$, so $\mathbf{G}\mathbf{1}_M = \mathbf{F}(\mathbf{F}^{\top}\mathbf{1}_M) = \mathbf{0}_M \in \mathbb{R}^M$ and $\lambda_1 = 0$. Since $\mathbf{F} \neq \mathbf{0}$, $\operatorname{tr}(\mathbf{G}) = \|\mathbf{F}\|_F^2 > 0$ implies $\lambda_M > 0$. Because $p > 0$ and $0 \leq \lambda_m^p \leq \lambda_M^p$ for $m \in \{2,\dots,M\}$, summing the $M-1$ active modes gives $\lambda_M^p \leq \sum_{m=2}^{M}\lambda_m^p \leq (M-1)\lambda_M^p$. Inverting and multiplying by $\lambda_M^p > 0$ reverses the chain to $\frac{1}{M-1} \leq R_{\text{LT}}(\mathbf{F}) \leq 1$, with equality at $\frac{1}{M-1}$ iff $\lambda_2 = \cdots = \lambda_M > 0$ and equality at 1 iff $\lambda_2 = \cdots = \lambda_{M-1} = 0 < \lambda_M$. Scaling $\mathbf{F} \mapsto \alpha\mathbf{F}$ ($\alpha \neq 0$) scales $\lambda_m \mapsto \alpha^2\lambda_m$ and cancels in (5). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

Rather than summing the 224 dials into a scalar norm, Stage 4 (Fig. 1, Algorithm 1) equips each dial with a binary circuit breaker calibrated on nominal data. Because all 224 dials derive from one 4×32 residual tensor, their trips are correlated by construction (median effective rank 65 of 224 on the development cohort, with joint trip-count variance $7.2\times$ independent trials), so alarm quorums are calibrated directly on the empirical distribution of simultaneous nominal trip counts:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1,\dots,224\}$ is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\,\text{MAD}(d_{j,\text{cal}}) + 10^{-4}$ into $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any individual coordinate that drifts during calibration, each dial's 98.5th-percentile tripwire τ_j is scaled by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\text{cal}}\,\gamma_j^{\text{sc}}$:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\,\gamma_j^{\text{cal}}\,\gamma_j^{\text{sc}}, \tag{6}$$
where γ_j^{cal} is the ratio of second-half to first-half calibration 98th percentiles whenever > 1.35 (1 otherwise), and γ_j^{sc} is $(m_j + s_j)/(1.5\, s_{\text{group},j})$ whenever > 1.25 (1 otherwise, relative to perspective-group median $s_{\text{group},j}$).
2) *32 fast right-edge onset relays:* To trigger rapidly when an abrupt fault enters the right edge of the S-step window while earlier prefixes ($m \in \{1,2\}$) still observe nominal data, each coordinate $c \in \{1,\dots,32\}$ defines a strict edge switch $b^{\text{edge}}_{p,c}(t) = \mathbb{I}\big(z_{p,c}(t) > q_{0.993}(z_{p,c,\text{cal}})\,\gamma^{1.25}_{p,c}\big)$ and fires its onset relay $r_c(t) \in \{0,1\}$ when a terminal-prefix trip ($m = 4$) coincides with acceleration, fluctuation, or prefix $m = 3$:
$$r_c(t) = b^{\text{edge}}_{\text{pos},4,c}(t) \wedge \big(b^{\text{edge}}_{\text{accel},c}(t) \vee b^{\text{edge}}_{\text{fluc},c}(t) \vee b^{\text{edge}}_{\text{pos},3,c}(t)\big). \tag{7}$$
3) *Causal trailing vs. buffered attribution and Boolean quorums:* Under *causal trailing attribution*, switches on window $[t, t+S)$ emit a per-timestep decision at its right endpoint $\tau = t + S - 1$ ($\mathcal{F}_\tau$-measurable, structural lag $S-1$, empirical delay $\Delta t = t_{\text{alm}} - t^*$). Under *retrospective buffered attribution*, the 224-dial vector routes position votes to $[t, t+S_m)$, drift and fluctuation votes to $[t, t+S)$, and endpoint-difference votes to $[t+S_3, t+S)$; the 128-dial counter and onset relays average uniformly over $[t, t+S)$ (at $S = 1$, the buffered endpoint-difference span is empty). Summing active breakers gives $N_{224}(\tau)$, $N_{128}(\tau)$, and $N_{\text{edge}}(\tau)$ (continuous ranking score $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ for PR-AUC). Per-timestep binary alarms $\hat{y}(\tau) \in \{0,1\}$ trigger when any counter exceeds its nominal-calibration quorum:
$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \;\vee\; N_{128}(\tau) > K_{128} \;\vee\; N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \tag{8}$$
where $K_{\text{edge}} = \max\big(1.25\, q_{0.999}(N_{\text{edge,cal}}),\, 2.5\big)$ and $(K_{224}, K_{128}) = \big(\max(1.12\, q_{0.995},\, 5.0),\, \max(1.12\, q_{0.995},\, 3.5)\big)$ on nominal trip counts.

E. Autopsy-Distilled Architecture: Laminar-JEPA-Lite

Layer-by-layer activation tracing and component ablations of *Laminar-JEPA* (Section VI) reveal four structural simpli-

fications that cut trainable parameters by 35%–52% (141.9k at $K < 25$; 222.3k at $K \geq 25$ vs. 295.8k) and accelerate windowed GPU inference by 2.43× (3.14× at $K < 25$) without post-training weight editing:

1) *Four-block TCN and $K \geq 25$ dyadic gating:* Because a fifth dilated block attenuates anomaly divergence on short target prefixes without raising effective rank (R_{stab}), *Laminar-JEPA-Lite* uses four dilated blocks ($d \in \{1, 2, 4, 8\}$), compacts its temporal/statistical pooling heads to 64D, and enables the four-shell rFFT front-end only when $K \geq 25$ (zero shell dropout; direct causal projection when $K < 25$).
2) *Parallel three-horizon single-LayerNorm predictor:* Replacing the sequential $M = 4$ rollout and redundant predictor LayerNorm stack with $M = 3$ nested prefixes ($S_m \in \{16, 32, 64\}$ at $S = 64$, 96×96 resolvent precision $\mathbf{\Omega} \in \mathcal{S}_+^{96}$) and a single parallel residual predictor, $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(z_{\text{ctx}} + \mathbf{p}_m + \mathbf{W}_2\, \phi_{\text{GELU}}(\text{LN}(\mathbf{W}_1(z_{\text{ctx}} + \mathbf{p}_m)))\big)$, removes sequential unrolling and LayerNorm cross-coupling.
3) *128-dial position-plus-drift panel and single quorum:* Retaining only the $3D = 96$ sub-horizon magnitudes $|\tilde{e}_{m,c}(t)|$ and $D = 32$ horizon-mean drift dials $|\bar{e}_c(t)|$ ($4D = 128$ breakers total, calibrated via (6)) eliminates redundant fluctuation/endpoint dials, onset relays, and post-hoc quenching, firing under one nominal quorum $\hat{y}_{\text{Lite}}(\tau) = \mathbb{I}\big(N_{128}(\tau) > \max(1.12\, q_{0.995}(N_{128,\text{cal}}), 4.0)\big)$ (prefix-weighted across $[t, t + S_m)$ in buffered mode).

1D Scalar Collapse Controls: To isolate panel gains over 1D scoring on identical backbones (Section VI), we compare against soft saliency pooling $g_d(\mathbf{v}) = (1 - \alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}$ ($w_i \propto \exp(|v_i|/(\tau_g \sigma_{\mathbf{v}}))$, $\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA scores $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$, and 1D Multi-Horizon JEPA scores $\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ (modulated by $1 + \alpha_{\text{LT}} \max(0, \tanh(0.5\, z_{\text{LT}}(t)))$ for $\alpha_{\text{LT}} \in \{0.12, 0\}$), calibrated via SPOT [8], [37], [51] ($u = p_{98}, q = 10^{-3}$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures use context $C = 256$, target horizon $S = 64$ ($W = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), latent dimension $D = 32$, nominal training stride 16, and evaluation stride 1. Models are trained with AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) and zero per-dataset tuning: batch 16 for 15–20 epochs on development cohorts, and batch 16–64 for 10–12 epochs with gradient clipping (1.0) and cosine annealing on archive cohorts.

a) Disjoint Nominal Calibration Protocol, Boundary Isolation, and Cross-Fitting: Each stream's nominal history $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$): (1) *Stage A (Weight fitting vs. checkpoint selection):* Standardization statistics, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and precision $\mathbf{\Omega} \in \mathcal{S}_+^{MD}$ are fit on $\mathbf{X}_{\text{tr}}$. Because monitoring early-stopping loss on nominal calibration windows can make empirical calibration quantiles mildly optimistic, all archive benchmarks ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$, Table V) enforce a strictly *disjoint checkpoint-selection protocol* that splits $\mathbf{X}_{\text{tr}}$ chronologically into weight-fitting $\mathbf{X}_{\text{fit}}$ ($\approx 50\%$ of T_{nom}) and checkpoint-selection $\mathbf{X}_{\text{sel}}$ ($\max(W + 64, \lfloor 0.15\, T_{\text{nom}} \rfloor)$ steps), leaving threshold calibration completely untouched by model training and selection (development cohorts train for fixed epochs on $\mathbf{X}_{\text{tr}}$ without validation early stopping). On short histories where carving out $\mathbf{X}_{\text{sel}}$ shrinks training support, a *two-fold chronological cross-*

Algorithm 1 Laminar-JEPA (224-Bit) and Laminar-JEPA-Lite (128-Bit)

Require: Nominal history $\mathbf{X}_{\text{nom}}$, test stream $\mathbf{X}_{\text{test}}$, tier $\in$ {Full, Lite}, attribution mode $\in$ {causal, buffered}.
Ensure: Per-timestep binary alarms $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$.

Phase 1: Disjoint Training & Resolvent Whitening
Split $\mathbf{X}_{\text{nom}}$ chronologically into $(\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_{\text{cal}}, \text{gap}_S, \mathbf{X}_{\text{hold}})$ and standardize channels using $\mathbf{X}_{\text{fit}}$ alone.
Set nested prefixes $S_{1:4} = \{16, 32, 48, 64\}$ (Full) or $S_{1:3} = \{16, 32, 64\}$ with $K \geq 25$ rFFT gate (Lite).
for stride-16 training batches in $\mathbf{X}_{\text{fit}}$ **do**
    Encode $z_{\text{ctx}} = E_\theta(\mathbf{x}_{\text{ctx}})$ and targets $z_{\text{tgt}}^{(m)} = E_\phi(\mathbf{x}_{\text{tgt}}^{(m)})$ with barycenter shift $-\bar{\boldsymbol{\mu}}_{\text{ctx}}$.
    Roll out sequential $\{P_{\text{sub}}, P_\theta\}$ (Full) or parallel single-LayerNorm predictor (Lite); update θ via (2) and $\phi \leftarrow m\phi + (1 - m)\theta$.
end for
Select checkpoint on $\mathbf{X}_{\text{sel}}$ (or fixed epochs) and fit $(\bar{\boldsymbol{\mu}}_{\text{res}}, \mathbf{\Omega} \in \mathcal{S}_+^{MD})$ on $\mathbf{X}_{\text{fit}}$.
if tier = Full and $q_{\text{wsp}} = 1$ **then**
    Rank coordinates by r_c (3) on $(\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{cal}})$; run Q1–Q4 for $k \in \{4, 2\}$ and keep edit iff nominal split and 0.92-quorum gates pass.
end if

Phase 2: Nominal Circuit-Breaker Calibration on $\mathbf{X}_{\text{cal}}$
Extract stabilized whitened residuals $\tilde{\mathbf{E}}_{\text{cal}}$ via (4).
if tier = Full **then**
    Expand $\tilde{\mathbf{E}}_{\text{cal}}$ into $7D = 224$ dials (position, drift, fluctuation, endpoint) and 32 onset relays (7).
    Fit MAD scales (m_j, s_j), hijack guards γ_j, tripwires τ_j (6), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (8).
else ▷ Distilled Laminar-JEPA-Lite
    Extract $4D = 128$ position-plus-drift dials $(|\tilde{e}_{m,c}|, |\bar{e}_c|)$, fit τ_j (6), and set single quorum $K_{\text{Lite}} = \max(1.12\, q_{0.995}, 4.0)$.
end if

Phase 3: Online Inference on $\mathbf{X}_{\text{hold}}$ or $\mathbf{X}_{\text{test}}$
for each stride-1 window $[t - C, t + S)$ **do**
    Compute $\tilde{\mathbf{E}}(t) \to \mathbf{d}(t)$ and binary trips $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$ (plus relays $r_c(t)$ if Full).
    Attribute active counts to right edge $\tau = t + S - 1$ (causal) or prefix spans $[t, t + S_m)$ (buffered).
    Emit $\hat{y}(\tau) \in \{0, 1\}$ when active counts exceed $(K_{224}, K_{128}, K_{\text{edge}})$ (8) or K_{Lite}.
end for

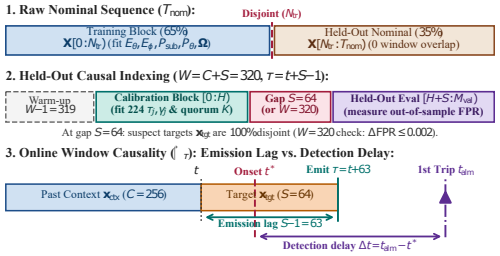


Fig. 2. Two-stage held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums $(K_{224}, K_{128}, K_{\text{edge}})$, and (Row 3) structural emission lag ($S-1$) versus empirical fault-detection delay (Δt).

fitting check swaps $[0 : H)$ and $[H + S : M_{\text{val}})$ across two calibration folds; on the 19-stream cache, consensus voting ($\hat{y}^{(1)} \wedge \hat{y}^{(2)}$) lowers stream-mean FPR_{test} from 1.50% to 0.92% (median $1.07\% \rightarrow 0.38\%$) while retaining 66.7% median ER. (2) *Stage B (Circuit-breaker calibration and boundary isolation on $\mathbf{X}_{val}$):* Stride-1 windows of length $W = 320$ inside $\mathbf{X}_{\text{val}}$ yield $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window indices. Splitting at $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and *held-out nominal evaluation block* $[H + S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H+S)$ so calibration and evaluation share zero target observations (a full $G = 320$ window gap changes causal FPR by < 0.25 percentage points). On $[0 : H)$ alone, the panel fits $w_{m,c}$ (4), dial medians m_j and MAD scales s_j, hijack guards γ_j, tripwires τ_j (6), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (8) (or single N_{128} quorum in *Laminar-JEPA-Lite*, τ_{SPOT} for 1D baselines), frozen on $[H + S : M_{\text{val}})$ and $\mathbf{X}_{\text{test}}$. We distinguish raw calibration window count $N_{\text{cal}} = H$ from autocorrelation-adjusted *effective sample size* $N_{\text{eff},s} = N_{\text{cal}}/\tau_{\text{int},s}$ using stream- and statistic-specific Sokal initial-positive-sequence estimators: across the 19 development streams, median τ_{int} is 2.51 (IQR $[1.52, 4.04]$) for individual dials z_j, 1.88 ($[1.22, 2.73]$) for N_{edge}, 5.58 ($[3.07, 14.09]$, max 206.0) for N_{224}, and 6.75 ($[5.10, 16.88]$) for 1D scalar scores (legacy lag-64 Bartlett pooled summary 2.66 median, 3.23 trimmed cohort summary; Section VII-E).

b) Hierarchical Statistical Units, Dataset-Clustered Inference, and Hypothesis Testing: Because streams within multi-stream archives share hardware platforms, we report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Dataset-clustered and stream-level inference:* Because stream-level p-values treat same-spacecraft channels as independent, *dataset-clustered inference* (95% cluster-bootstrap CIs and exact 2^D dataset sign-flip permutation tests) is emphasized alongside stream-level paired Wilcoxon tests and 95% stream-bootstrap CIs (2,000 resamples). For multiplicity reporting on the 118 untouched streams, the focal causal contrasts comparing *Laminar-JEPA-Lite* against Multi-Horizon JEPA (1D) and TS-JEPA yield Benjamini–Hochberg [40] adjusted stream-level $p = 0.0034$ on Point-F1 (raw $p = 0.0034$ and 1.6×10^{-4}) and adjusted dataset sign-flip $p = 0.039$ on ER (raw $p = 0.012$ and 0.039, positive on $8/9$ datasets); other quoted p-values are raw and descriptive. Non-Exathlon held-out subsets ($n = 101$, 8 datasets) verify insensitivity to archive composition. (2) *12-fold LODO audit:* Across all $12/12$ LODO folds on the 141-stream archive, *Laminar-JEPA-Lite* retains a positive causal Point-F1 gain over Multi-Horizon JEPA (1D) ($\Delta \in [+0.002, +0.039]$, $\max p = 0.015$) and TS-JEPA ($\Delta \in [+0.023, +0.034]$, $\max p = 3 \times 10^{-4}$), and selecting the detector maximizing causal Point-F1 subject to $\text{FPR}_{\text{hold}} \leq 6\%$ on $D - 1$ datasets chooses *Laminar-JEPA-Lite* in all $12/12$ folds.

B. Consolidated Cohort Organization, Development vs. Held-Out Separation, and Baseline Fairness

Across all evaluations, *Laminar-JEPA* denotes the sequential four-horizon 224-bit architecture with nominal split-gated quenching (*Laminar-JEPA (w/o Quench)* omits Steps Q1–Q4), and *Laminar-JEPA-Lite* denotes the parallel three-horizon 128-bit position-plus-drift architecture without weight editing. Both causal trailing and retrospective buffered attribution are paired on identical splits within each cohort (aside from a single-horizon 130-stream buffered baseline screen in Section VII-A): (1) *Development cohorts (design-inspected):* A recomputed 19-stream residual cache across 6 datasets evaluates the dial ladder, component ablations, and conformal/SPC comparators on identical 4H unquenched residuals (Table I(a,b), Table II); a separately trained 33-stream development benchmark across 7 datasets (19 non-GHL matched-backbone streams in Table I(c) plus 14 GHL streams in Table III) pairs *Laminar-JEPA*, *Laminar-JEPA-Lite*, 1D JEPAs, and observation-space baselines; and tripwire/quenching constants were fixed on 23 tuning streams (19 non-GHL plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`). (2) *Untouched held-out cohorts (primary generalization evidence):* Excluding those 23 design streams from the 141-stream archive (12 datasets, strictly disjoint $\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$) leaves 118 **untouched held-out streams** across 9 datasets (101 non-Exathlon streams across 8 datasets; Table V), supplemented by a single-seed 4H Q1–Q4 step-isolation audit on the same 118 streams (Section VI-A). None of the 118 held-out streams was inspected during architecture or threshold design.

Standardized Baseline Fairness Protocol: Scalar baselines (TS-JEPA, Multi-Horizon JEPA (1D), TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [16]) share the identical chronological splits, training-only standardization, isolation gaps, and SPOT calibration ($u = p_{98}$, $q = 10^{-3}$), whereas conformal and SPC detector-bank baselines in Table II share the identical multi-horizon whitened residuals and nominal calibration alert rate. *Laminar-JEPA-Lite* uses 141.9k trainable parameters at $K < 25$ and 222.3k at $K \geq 25$ versus 295.8k for *Laminar-JEPA* (206k–649k across external neural baselines; circuit-breaker panels add zero trainable weights). Runs used PyTorch 2.4 / CUDA 12.4 on an RTX 5090 (0.15 ms/window batched inference for *Laminar-JEPA-Lite* vs. 0.38 ms/window for *Laminar-JEPA*, $2.43\times$ speedup; 188 MB peak memory); code: https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics and Aggregation

To prevent point-adjustment inflation [41] and aggregation ambiguity, all binary false-alarm and Point-F1 metrics are computed *per valid timestep* τ (never point-adjusted and never per-window without timeline attribution): FPR_{hold} averages $\hat{y}(\tau)$ over valid timesteps in $[H + S : M_{\text{val}})$, FPR_{test} averages $\hat{y}(\tau)$ over nominal test timesteps ($y(\tau) = 0$), and Point-F1, Precision, Recall, and PR-AUC (from $N_{\text{tot}}(\tau)$) score all valid test timesteps against $y(\tau) \in \{0, 1\}$. Event-level ER measures the fraction of contiguous $y = 1$ segments with ≥ 1 alarm, and delay $\Delta t = t_{\text{alm}} - t^*$ is the step lag from segment onset t^* to first in-segment alarm. Unless explicitly labeled

median or *Dataset-Macro*, cohort headline numbers report *stream-level arithmetic means* of within-stream seed averages, paired with stream medians for skewed FPR distributions and non-deployable post-hoc test-FPR controls ($f^* \in \{0.01, 0.05\}$) strictly for diagnostic comparison.

VI. Ablation Studies and Analysis

A. Ablation 1: Perspective Expansion Ladder, Circuit-Breaker Components, and Weight-Space Quenching

Table I(a),(b) and Fig. 3(a) evaluate perspective expansion and panel components on the 19-stream residual cache (not paired to the separately trained run in Panel (c)):

1) *Operational trade-off between 160 and 224 dials:* While the 224-dial panel lifts causal Point-F1 from 0.013 to 0.071 over 1D scoring ($p = 0.0023$; buffered $+0.033$, $p = 0.53$), more dials are not automatically better: the 160-dial position-plus-drift subset (128 sub-horizons + 32 horizon means, single quorum, no relays) achieves higher causal Point-F1 (0.098), higher causal ER (59.1% vs. 56.5%), shorter causal delay (74.8 vs. 90.1 steps), and lower nominal FPR (0.55% held-out, 0.87% test vs. 0.76%, 1.50%) at 29% lower dial cost, whereas all 224 dials lead only in retrospective buffered Point-F1 (0.224 vs. 0.198). This directly motivates the $M = 3$ position-plus-drift 128-dial panel in *Laminar-JEPA-Lite*. Scalar R_{LT} modulation remains null in Panel (c).
2) *Relays, single-quorum voting, and hijack guards:* Removing the 32 onset relays lowers causal Point-F1 slightly ($0.071 \to 0.068$), and replacing the three-counter rule (8) with a single 224-dial quorum lowers causal Point-F1 to 0.063 ($\text{FPR}_{\text{hold}} = 0.0055$; Table II). Conversely, removing hijack guards ($\gamma_j = 1$) raises causal Point-F1 to 0.073 and shortens delay to 70.1 steps on this cache, showing that drift guards can damp useful fault responses when calibration drift is mild. None removes the $S - 1$ causal reporting lag.
3) *Quenching step isolation and LayerNorm coupling:* On single-seed 4H checkpoints across all 118 untouched streams (unquenched causal $\text{FPR}_{\text{hold}} = 6.76\%$, F1 $= 0.0901$, ER $= 63.8\%$), $k = 4$ predictor row damping (Step Q1) shifts unselected coordinates by a median 11.1% of nominal RMS (45.5% shifting $> 10\%$) via 32D LayerNorm coupling without lowering FPR_{hold} (7.06%), while horizon-embedding shifts lower raw RMS on 68.2% of edited coordinates (4.1% spillover). Buffer recentering and precision decoupling alone (Steps Q2–Q3) drive the FPR drop ($\text{FPR}_{\text{hold}} = 6.71\%$, $\text{FPR}_{\text{test}} = 14.75\%$) at lower causal F1 ($0.0826$), whereas adding the nominal split gate (Step Q4, accepted on $39/118$ streams) reaches $\text{FPR}_{\text{hold}} = 6.62\%$ while preserving causal F1 (0.0904, ER $= 63.3\%$). Because ungated edits lower $k = 2$ causal F1 on $11/19$ development streams ($0.141 \to 0.049$) and $k \in \{8, 12\}$ cuts PR-AUC ($0.292 \to 0.223$), *Laminar-JEPA-Lite* omits weight editing altogether.

Conformal, multiplicity-controlled, and SPC detector banks on identical residuals: Table II benchmarks the panel against split-conformal, multiple-testing, and SPC detectors on the same 19-stream multi-horizon residuals without test-label tuning. Under family-wise Šidák/Bonferroni or Benjamini–Hochberg (0.5%) control across 128 coordinates or 224 dials, discrete calibration p-values have minimum resolution $(N_{\text{cal}} + 1)^{-1} > 3.9 \times 10^{-5}$, paralyzing causal recall (F1 ≤ 0.004). Matching per-coordinate conformal thresholds to the panel's nominal calibration alert rate (0.34%) causes tie-induced false-alarm inflation ($\text{FPR}_{\text{hold}} = 21.1\%$), as do recalibrated Mahalanobis conformal, Hotelling T^2, MEWMA, and CUSUM charts (27.9%–58.6%). By contrast, training-whitened multivariate ℓ_2 split-conformal scoring (F1 $= 0.072$, ER $= 60.8\%$, $\text{FPR}_{\text{hold}} = 0.62\%$), Fisher/Stouffer p-value pooling ($0.109/0.095$), and the 160-dial subset (0.098) match or exceed the 224-dial panel on this cache, demonstrating that training-fixed whitening and bounded coordinate aggregation—not 224-dial complexity alone—drive false-alarm stability.

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

Even when flatlining shifts the normalized summary (0.64 vs. 0.77 relative divergence for a variance-matched harmonic flip on `SMAP P-3`), terminal LayerNorm spreads static error thinly across all 32 coordinates (12.5 vs. 62.1 switches; 6.3% vs. 97.7% window alerts), and single-sensor faults suffer further $O(1/K)$ dilution across $K - 1$ nominal channels. Extending Fig. 3(b) (n=6), we evaluate a causal channel branch across all 19 multivariate development streams ($K \in [5, 55]$) without per-stream or test-label tuning. Per channel, three 64-step rolling statistics—log-variance drop $-\ln(\text{Var}_{64}(x_c) + 10^{-4})$, training-fit first-order autoregressive (AR(1)) residual RMS, and mean shift $|\bar{x}_{64,c} - \mu_c|/\sigma_c$—are thresholded at $1.25\, q_{0.999}$ on nominal validation windows (floor 6.0 MAD, $1.25\times$ half-split drift multiplier) with a 24-of-40-step persistence gate. On the highest-variance active channel $c^\star = \arg\max_{c:\sigma_c^{\text{tr}} > 10^{-8}} \sigma_c^{\text{tr}}$, a deterministic 9-point grid over $[320, N_{\text{test}} - 520]$ (stride-50 fallback) places three nonoverlapping $L_{\text{inj}} = 200$-step faults (57 events/class, gap ≥ 264) inside clean segments ($y^{\text{orig}} \equiv 0$ on $[s - 64, s + 264)$). For $s \leq t < s + 200$ using training $(\mu_c, \sigma_c, q_{.995,c})$:

$$x'_{t,c^\star} = \begin{cases} \mu_{c^\star} & \text{variance collapse,} \\ \mu_{c^\star} + 1.35(-1)^{t-s}\sigma_{c^\star} & \text{prediction shock,} \\ x_{s-1,c^\star} & \text{sensor freezing,} \\ \max(q_{.995,c^\star}, \mu_{c^\star} + 2.2\sigma_{c^\star}) & \text{saturation,} \\ x_{t,c^\star} + 2.5\sigma_{c^\star}(t-s)/199 & \text{gradual degradation,} \\ x_{t,c^\star} + 1.5\sigma_{c^\star} & \text{isolated bias.} \end{cases} \tag{9}$$

Outside $[s, s + 200)$, all channels remain unchanged. Injected-event recall rises from 63.2% (standalone panel) to 93.0% for variance collapse, 63.2% to 94.7% for freezing, and 82.5% to 94.7% (96.5% full branch) for saturation under the variance-collapse union; the full channel branch raises recall from 75.4% to 96.5% on prediction shocks, 59.6% to 80.7% on gradual degradation, and 68.4% to 86.0% on isolated bias. However, on the *unmodified* test streams, causal FPR_{test} rises from 1.5% (standalone panel, F1 $= 0.071$) to 6.8% (variance-collapse union, F1 $= 0.070$) and 19.4% (full branch, F1 $= 0.129$). Synthetic single-channel injections do not establish detection on natural sensor failures, and the full

TABLE I
CIRCUIT-BREAKER DIAL EXPANSION, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY

(a) Perspective Expansion Ladder ($1 \rightarrow 224$ Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Horizon Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Horizon Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (4). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Laminar-JEPA (224-Bit)	0.92	0.190	0.013	0.128	0.144	**0.244**	**0.311**	**0.407**	0.673	0.795	0.612	0.757
Laminar-JEPA (w/o Quench)	0.88	0.190	0.016	0.128	0.142	0.235	0.304	0.381	**0.695**	0.815	**0.631**	0.765
Laminar-JEPA-Lite (128-Bit)	2.33	0.182	0.010	0.061	0.133	0.242	0.301	0.397	0.650	0.811	0.560	**0.772**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Multi-Horizon JEPA (1D)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Multi-Horizon JEPA (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

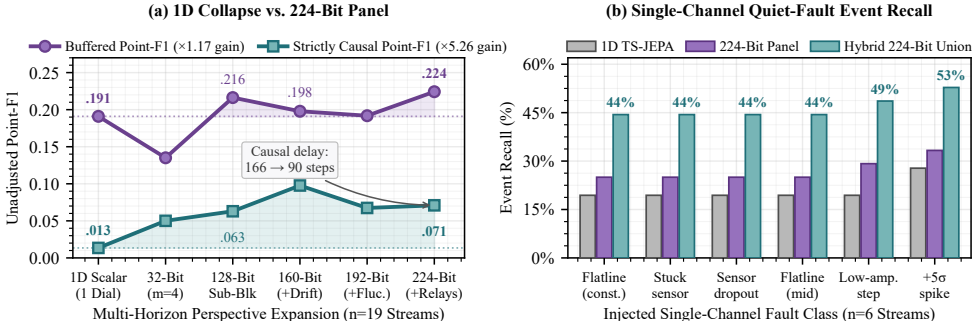


Fig. 3. Development studies. **(a)** Causal and buffered Point-F1 across the six-step expansion on cached whitened residuals from 19 streams (three seeds; no stabilization blend). **(b)** Event recall for injected single-channel faults in a separate six-stream study, comparing 1D TS-JEPA, 224-Bit Breaker, and their channel-level union.

TABLE II
CONFORMAL AND SPC COMPARATORS ON CACHED RESIDUALS

Detector (Calibration Rule)	Pt-F1 ↑	ER ↑	FPR_{hold} ↓	FPR_{test} ↓
224-dial panel (deployed)	0.071	0.565	0.007,6	0.015,0
160-dial P+D (single quorum)	0.098	0.591	0.005,5	0.008,7
224-dial single quorum (no relay)	0.063	0.564	0.005,5	0.008,8
Per-coord. conformal (Šidák .005)	0.000	0.000	0.000,0	0.000,0
224-dial BH FDR (q=.005)	0.004	0.018	0.000,0	0.000,1
Per-coord. conformal (matched)	0.265	0.843	0.211,1	0.283,1
Multivar. conf. ($\|\mathbf{e}_{\text{white}}\|_2^2$, matched)	0.072	0.608	0.006,2	0.009,5
Multivar. conf. (Mahal., α=.005)	0.279	0.896	0.278,9	0.473,9
Fisher p-pool (matched)	0.109	0.591	0.010,5	0.018,3
Maximum dial z (matched)	0.083	0.565	0.009,3	0.015,7
Hotelling T^2 (matched)	0.283	0.896	0.290,1	0.483,1

19 streams × 3 seeds, causal means. Matched rules use each stream's nominal panel alarm rate (mean 0.34%; Stouffer F1 .095, FPR_{hold} .0087; Top-k F1 .070, FPR_{hold} .0077; MEWMA .4301, CUSUM .5862 FPR_{hold}).

union's 19.4% false-alarm rate makes it unsuitable as a default detector.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining the multi-horizon JEPA backbone with and without 4-shell rFFT splitting on five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, `GECCO`; three seeds) reveals a clear channel-count threshold: shell splitting improves high-channel spacecraft detection (K=55 `MSL M-1` buffered F1 0.085 → 0.410; K=25 `SMAP P-3` causal F1 0.212 → 0.321) by separating thermal drift from actuator chatter, but lowers F1 at K=9 (`Daphnet`, `GECCO`), setting the $K \geq 25$ spectral gate in *Laminar-JEPA-Lite*.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 16 matched-budget arms in Table I(c) isolates five mechanisms: (1) raw-observation forecasting and autoencoding share heavy tails ($\gamma_2 \approx 6.2$), large drift ($D_{\text{KS}} \approx 0.45$), and ≈ 25% causal FPR; (2) a D=32 LayerNorm bottleneck in observation space cuts kurtosis ($\gamma_2 = 0.91$) and halves FPR, yet raw targets keep drift high ($D_{\text{KS}} = 0.376$); (3) learned D=32 latent targets and whitening halve drift ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179–0.190 across multi-horizon JEPA tiers, $p \leq 0.001$); (4) linear PCA and Isolation Forest keep low FPR but miss nonlinear faults; and (5) frozen targets and ℓ_2 scoring retain $< 2\%$ causal FPR, whereas latent context autoencoding lowers causal F1.

VII. Results and Discussion

A. Development 33-Stream and Untouched Held-Out 118/141-Stream Benchmark Results

Following the cohort hierarchy in Section V-B, Table III reports the 33-stream development benchmark under retrospective buffered attribution (Panel (a)), causal streaming (Panel (b)), and per-dataset stratification (Panel (c)), while Table V reports the strictly disjoint-selection ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$) benchmark across all 118 untouched held-out streams and the 141-stream archive. On a separate 130-stream baseline screen, TS-JEPA reaches stream-mean Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082. The observed reduction in nominal false alarms with latent scoring is specific to these protocols; spherical normalization alone does not prove a distributional guarantee.

Operational trade-offs and configuration guidance: On the 33-stream benchmark, *Laminar-JEPA* edges out *Laminar-JEPA-Lite* on unweighted buffered Point-F1 (0.244 vs. 0.242) and causal ER (67.3% vs. 65.0%), whereas *Laminar-JEPA-Lite* halves buffered FPR_{hold} (6.1% vs. 12.8%), leads 7-dataset macro Point-F1 (0.171 vs. 0.133), and cuts parameters by 35%–52% and inference latency by 2.43×. In practice, *Laminar-JEPA-Lite* is the recommended default for streaming and buffered deployment, *Laminar-JEPA* suits settings prioritizing peak buffered F1 or nominal 4H drift damping, short horizons ($S \leq 4$) address sub-window reporting lag, and raw-level variance or trend charts should be paired only when targeting single-sensor freezes or slow monotonic process ramps.

Scope boundaries and controlled slow-ramp sweep: Observation-space reconstructors show high nominal false-alarm rates here, whereas learned latent targets lower false alarms without guaranteeing light tails on arbitrary data. Conversely, because the encoder summarizes only the preceding $C = 256$ steps, gradual drift that shifts negligibly across one context window causes context and target summaries to co-move, explaining the 0.000 Point-F1 on the 14 GHL slow-ramp streams (low false alarms without recall is not detection). To map this boundary, we injected linear ramps of amplitude $a \in \{0.5, 1, 2, 4\}\sigma$ and duration $L \in \{64, 256, 1024, 4096\}$ (slopes a/L from 6.25×10^{-2} to $1.22\times10^{-4}\sigma$/step, plus a 256-step plateau; 15 events/cell across 10 clean test segments on $\lceil 0.3K \rceil$-channel and 1-channel subsets). For steep multi-channel ramps ($L = 64$, $a \in [1, 4]\sigma$), the 224-bit panel reaches 86.7%–88.7% causal detection probability at 46–54-step median delay and 1.7%–1.9% false-alarm rate (+5.7–7.7 percentage points above the clean-segment Poisson episode baseline; +19.4–24.1 points for 1D SPOT at 60–67-step delay). By $L = 256$ ($= C$), panel excess causal detection narrows to +0.5–2.5 points (50–71-step delay), and for $L \geq 1024$ (slope $\leq 3.9\times10^{-3}\sigma$/step) raw hit rates (87%–91%) reflect background episodes over 1,280–4,352-step spans, with chance-corrected detection falling to $[-4.8, +0.8]$ points across all JEPA detectors ($[-5.9, -1.0]$ on single-channel ramps). Fixed-window JEPAs thus target abrupt to medium-onset faults rather than gradual industrial drift.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table III shows why 1D latent detectors degrade under real-time causal streaming. Buffered attribution averages residuals across overlapping predictions covering each timestamp ($\gamma_2 = 1.25$), whereas causal streaming evaluates a single right-edge step without temporal averaging ($\gamma_2 = 0.92$–2.33). Summing whitened coordinates into a 1D score allows one or two drifting coordinates to inflate the global SPOT threshold onto that heavier tail, missing sustained faults. Replacing the scalar sum with independently calibrated coordinate breakers isolates drifting coordinates. Against a rate-matched Bernoulli alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, random guessing catches only 28.1% of events at our $\leq 1.3\%$ causal FPR, whereas at TimesNet's 24.3% causal FPR, coin flipping catches 60.0% of events by chance alone, exceeding TimesNet's 58.8% recall.

C. Multi-Horizon Scaling Across $S \in \{1, 4, 16, 32, 64\}$ and Latency Trade-Offs

Because a target window of length S emits its causal decision at $\tau = t + S - 1$, $S = 64$ imposes a 63-step reporting lag. We retrain on five selected streams at $S \in \{1, 4, 16, 32, 64\}$ to examine the latency trade-off. At $S = 1$, all predictor states are compared with the same one-step target, so this setting has no distinct target-horizon decomposition; at $S = 4$, the shortest encoded prefix is two steps. In the buffered $S = 1$ setting, endpoint-difference votes have no routed span. These short-horizon results therefore test the implemented detectors, not distinct horizon views.

In Fig. 4(a), both *Laminar-JEPA* and *Laminar-JEPA-Lite* exceed 1D Multi-Horizon SPOT across all five horizons while holding test FPR below 3.6% ($\leq 2.4\%$ for *Laminar-JEPA-Lite*, held-out FPR $\leq 1.2\%$, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). At $S = 1$ and $S = 4$, *Laminar-JEPA* reaches 2.6× and 4.3× the scalar causal Point-F1 (0.037 and 0.050 vs. 0.014 and 0.012) with 77%–84% ER, while *Laminar-JEPA-Lite* reaches 2.1× and 3.1× (0.030 and 0.036) with 64%–82% ER; at $S = 4$, detection delay drops to 36.7–44.1 steps. Buffered Point-F1 peaks at 0.290 for *Laminar-JEPA* ($S = 32$) and 0.255 for *Laminar-JEPA-Lite* ($S = 64$) versus 0.182 for 1D SPOT. Five selected streams cannot establish a deployment rule across sampling rates or datasets.

D. Operating Budgets and Contextual vs. Point Anomalies

Fig. 4(b) and Table IV examine detection across operating budgets and fault taxonomies:

1) *Autonomous vs. diagnostic operating budgets (Fig. 4(b), Table I(c)):* Along the causal recall–FPR frontier, the autonomous points of *Laminar-JEPA* and *Laminar-JEPA-Lite* occupy the $\leq 1.3\%$ held-out FPR regime without test-label tuning. At post-hoc diagnostic test-FPR controls ($f^* \in \{1\%, 5\%\}$), *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% (63.1% unquenched) and 56.0% of events at

TABLE III
PRIMARY 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-horizon coordinate circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.128	**0.244**	**0.407**	**0.669**	432.9	0.013	0.144	**0.673**	249.9	0.004	0.000	0.133
Laminar-JEPA-Lite (128-Bit)	0.061	0.242	0.397	0.636	59.3	0.010	0.133	0.650	73.8	0.005	0.000	**0.171**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Multi-Horizon JEPA (1D)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Multi-Horizon JEPA (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.214	0.179	0.145	**0.229**	0.144	**.009/.061**	.012/.091	.030/.091	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.003/.004**	.005/.005	.004/.004	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.078	**0.191**	0.000	0.000	0.000	.009/.104	.008/.020	.004/.017	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.284**	0.210	0.161	0.158	0.151	.013/.151	**.004/.049**	.054/.095	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.357	0.375	**0.379**	0.289	0.167	.021/.223	.016/**.066**	**.014**/.135	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.010/.014	.001/**.000**	.008/.014	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.000	**0.240**	0.119	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), trained separately from Table I(a,b); best in **bold**, second underlined. In (c), n = streams, K = channels, E = events, π = anomaly rate, $F1_{all} = 2\pi/(1+\pi)$, Lam. = Laminar-JEPA, Lite = Laminar-JEPA-Lite. Unweighted Laminar-JEPA Δt reflects one late SWAN trigger (0.0002 recall; median Δt is 50.5 buffered / 45.8 causal, or 54.8 / 57.2 ex-SWAN). 95% stream-bootstrap CIs (n=19): Lite buffered FPR [.030, .103], F1 [.151, .335] and causal FPR [.006, .014], F1 [.055, .236]; Lam. buffered F1 [.150, .357] and causal FPR [.009, .018], F1 [.058, .257] (vs. Multi-Horizon 1D [.013, .097] and TimesNet causal FPR [.123, .375]).

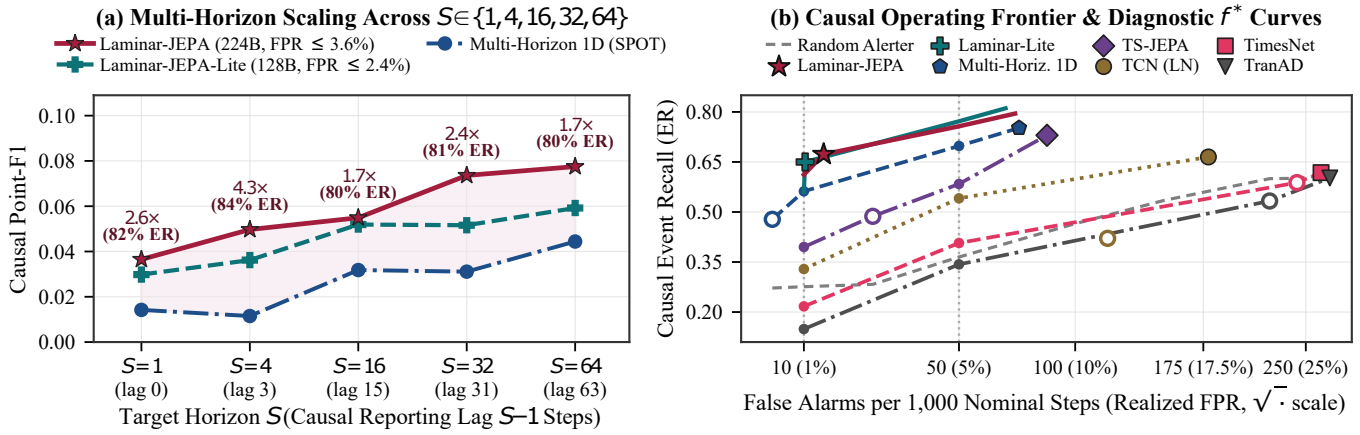


Fig. 4. Multi-horizon scaling ($S \in \{1, 4, 16, 32, 64\}$, causal reporting lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, n=5 streams) and causal operating frontier (n=19 streams). **(a)** Causal Point-F1 and event recall (ER) across S comparing *Laminar-JEPA (224B)*, *Laminar-JEPA-Lite (128B)*, and Multi-Horizon 1D SPOT (nominal FPR $\leq$ 3.6% across all S, vs. 2.5% $\rightarrow$ 21.3% for TimesNet). **(b)** Causal ER vs. realized false alarms ($\sqrt{\cdot}$ scale) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted guides), autonomous points (*Laminar-JEPA* star, *Laminar-JEPA-Lite* cross, 1D open circles), and quantiles $q^* \leq 0.05$.

$f^* = 1\%$ versus 39.5% for TS-JEPA, 21.7% for TimesNet, and 14.9% for TranAD.

2) *In-range contextual vs. point faults (Table IV):* Under NASA's spacecraft taxonomy [4] (Panel (a)) and a 5σ amplitude split across all 19 development streams (Panel (b)), out-of-range point spikes ($> 5\sigma$) are flagged by reconstructors and amplitude rules alike, whereas in-range contextual faults decouple cross-channel phase or velocity inside normal amplitude bounds: on NASA contextual sequences, *Laminar-JEPA-Lite* and *Laminar-JEPA* achieve 87.5% and 77.5% autonomous ER at 1.3% and 2.2% FPR (84.0% and 80.0% at $f^* = 1\%$; 0.532 and 0.519 buffered PR-AUC) versus 62.5% at 41.5% FPR (56.0% at $f^* = 1\%$, 0.317 PR-AUC) for TimesNet.

E. Untouched Held-Out Generalization, Calibration Sample-Size Reliability, and Drift Decomposition

Table V and archive-wide audits establish three generalization and calibration findings:

1) *Independent generalization under disjoint checkpoint selection (Table V):* Across the 118 untouched streams ($\mathbf{X}_{sel} \perp \mathbf{X}_{cal}$, three seeds), *Laminar-JEPA-Lite* raises causal Point-F1 from 0.074 (Multi-Horizon 1D) and 0.060 (TS-JEPA) to 0.085 (69/39 wins, $p = 0.0034$; +0.025 vs. TS-JEPA, cluster CI $[+0.005, +0.046]$, $p = 1.6 \times 10^{-4}$). It lifts buffered Point-F1 from 0.138 to 0.153 ($p = 0.0067$; vs. 0.141 for *Laminar-JEPA*, $p = 0.026$, cutting median buffered FPR_{hold} from 4.74% to 1.96%), raises causal ER from 49.5% to 67.0% across 8/9 datasets (cluster CI $[+10.5\%, +27.3\%]$, sign-flip $p = 0.012$; *Laminar-JEPA*

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION

(a) Model	NASA Contextual (n=5 seqs) Buf. PR	ER	FPR	f^*=.01	f^*=.05	NASA Point (n=8) f^*=.01	f^*=.05
Laminar-JEPA	0.519	0.775	0.022	0.800	**0.960**	0.783	0.900
Laminar-JEPA-Lite	0.532	**0.875**	0.013	**0.840**	0.920	0.700	0.967
Multi-Horiz. 1D	**0.652**	0.750	**0.006**	0.820	0.920	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

(b) Model	19-Stream Contextual ($\|z\|\leq 5$) q^*=.05	f^*=.01	f^*=.05	19-Stream Point ($\|z\|>5$) q^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	**0.669**	0.448	**0.650**	0.835	0.644	0.795
Laminar-JEPA-Lite	0.655	0.376	0.567	**0.848**	0.581	**0.801**
Multi-Horiz. 1D	0.628	0.498	0.590	0.782	0.683	0.781
TS-JEPA	0.614	**0.520**	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Event recall (ER; autonomous vs. diagnostic f^*); best in **bold**, second underlined. (a) 10 SMAP/MSL streams by NASA taxonomy [4] (contextual: E-1, A-2, A-7, M-1, C-1 [2100, 2210]; point: A-1, P-3, D-3, D-14, D-15, C-1 [550, 750]). (b) n=19 non-GHL streams split into in-range ($\max|z| \leq 5\sigma$) vs. out-of-range ($\max|z| > 5\sigma$) causal events.

TABLE V
DISJOINT-SELECTION HELD-OUT ARCHIVE BENCHMARK

(a) Untouched n=118	Pt-F1 ↑ Caus.	Buf.	PR-AUC ↑ Caus.	Buf.	ER ↑ Caus.	Buf.	FPR_{hold} ↓ Caus.	Buf.
TS-JEPA (1D)	.060	.131	.153	.200	.423	.475	**.042**	**.109**
Multi-Horizon JEPA (1D)	.074	.138	**.169**	**.234**	.495	.552	.059	.178
Laminar-JEPA (w/o Quench)	.077	.146	.147	.229	.639	.626	.056	.181
Laminar-JEPA (224B)	.073	.141	.144	.222	.621	.613	.054§	.174
Laminar-JEPA-Lite (128B)	**.085**†	**.153**‡	.148	.220	**.670**¶	**.686**	.057	.120
All Archive n=141 (12 **Datasets**)								
TS-JEPA (1D)	.055	.136	.165	.213	.426	.475	**.036**	**.103**
Multi-Horizon JEPA (1D)	.070	.146	**.182**	**.246**	.492	.543	.050	.163
Laminar-JEPA (224B)	.069	.150	.159	.237	.634	.623	.046	.161
Laminar-JEPA-Lite (128B)	**.084**	**.155**	.161	.233	**.665**	**.679**	.050	.107

(b) Held-out	n	Causal Pt-F1 ↑ Lite	Lam.	1D	Buffered Pt-F1 ↑ Lite	Lam.	1D	Causal FPR_{hold} ↓ Lite	Lam.	1D
SMAP	40	.097	.077	**.141**	.177	.175	**.201**	**.048**	.062	.125
SMD	18	.052	**.074**	.027	.124	**.135**	.067	.016	.031	**.004**
Exathlon	17	**.178**	.138	.085	**.217**	.168	.131	.181	.113	**.098**
Daphnet	15	**.088**	.070	.037	**.150**	.138	.093	.081	.075	**.009**
MSL	13	.023	**.032**	.026	.104	.112	**.211**	.010	.013	**.004**
OPPORT.	12	**.029**	.016	.010	**.093**	.048	.043	.011	.008	**.004**

Three seeds under $\mathbf{X}_{\text{sel}} \perp \mathbf{X}_{\text{cal}}$; best in **bold**, second underlined (1D = Multi-Horizon JEPA 1D). †Lite vs. 1D (n=118): causal F1 Δ=+0.012 (69/39 wins, p=0.0034; vs. TS-JEPA +0.025, cluster CI [+0.005, +0.046], $p=1.6\times10^{-4}$); positive in 12/12 LODO folds on n=141 ($\max p$=0.015). ‡Buf. F1 Δ=+0.015 (p=0.0067; vs. Lam. +0.012, p=0.026, median buf. FPR_{hold} 1.96% vs. 4.74%). ¶Causal ER Δ=+0.175 vs. 1D (cluster CI [+0.105, +0.273], 8/9 sets, sign-flip p=0.012; Lam. +0.126, p=0.016). §Quenching vs. unq. 4H: causal FPR_{hold} Δ=−0.002 (7/9 sets improved, 0 worse, sign-flip p=0.016, Wilcoxon $p=2.7\times10^{-4}$; median 0.96% vs. 1.25%).

62.1%, $p = 0.016$), and halves causal delay (62.7 vs. 141.9 steps). Excluding Exathlon ($n = 101$, 8 datasets), *Laminar-JEPA-Lite* holds causal FPR_{hold} at 3.7% (median 0.84%) and lifts causal ER from 47.7% to 63.5%; across all 141 streams (12 datasets), it stays positive in 12/12 LODO folds against Multi-Horizon 1D ($\max p = 0.015$) and TS-JEPA ($\max p = 3 \times 10^{-4}$). Nominal split-gated quenching in *Laminar-JEPA* lowers 4H causal FPR_{hold} across 7/9 datasets with zero worsened (5.57% → 5.37%, median 1.25% → 0.96%, sign-flip $p = 0.016$, Wilcoxon $p = 2.7 \times 10^{-4}$) while moving causal Point-F1 from 0.077 to 0.073.

2) *Autocorrelation and calibration sample size:* Across the 19 development streams, Sokal autocorrelation τ_{int} spans median 2.51 (IQR $[1.52, 4.04]$, max 63.5) for dials z_j, 1.88 ($[1.22, 2.73]$, max 79.0) for N_{edge}, 5.58 ($[3.07, 14.09]$, max 206.0) for N_{224}, and 6.75 ($[5.10, 16.88]$) for 1D scores ($\tau_{\text{int}} \approx 3.23$ is only a trimmed summary). At median $N_{\text{cal}} = 408$, median N_{eff} (expected tail count) is 154.6 (2.32) at $q_{0.985}$ for z_j, 67.1 (0.34) at $q_{0.995}$ for N_{224}, and 195.9 (0.20) at $q_{0.999}$ for N_{edge}; even at $N_{\text{eff}} = 300$, expected tail counts $(4.5, 1.5, 0.45, 0.3)$ do not establish precise extreme-quantile estimation. Moving-block bootstrap ($L_{\text{blk}} \in \{1..128\}$) relative 95% CI half-widths span 12.7%–17.5% on z_j and 19.5%–25.0% on $(N_{224}, N_{\text{edge}})$, where the 1.12× margin and floors $(5.0, 2.5)$ hold median FPR_{hold} to 0.36% and 0.00%. On the 118 untouched streams, $N_{\text{cal}} < 250$ ($n = 12$) averages 16.1% causal FPR_{hold} for *Laminar-JEPA-Lite* (17.7% *Laminar-JEPA*) versus 1.35% (2.56%) for $N_{\text{cal}} \geq 2{,}500$ ($n = 22$); Harrell–Davis Beta quantiles (1.26% vs. 1.67% exceedance at $N_{\text{cal}} = 250$) and IQR extrapolation remain exploratory.

3) *Regime drift vs. tail misfit via permutation decomposition:* Refitting thresholds across chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits is consistent with inter-block regime drift without ruling out parametric tail misfit: $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$ represents 71.3% of excess causal FPR in TS-JEPA and 78.3%–81.8% in reconstructors, versus 18.7% in the centered model. Within-checkpoint traces of *Laminar-JEPA* (n=6, $K \in \{9, 25, 55\}$) show pre-tap vectors clustering on an off-center cap ($\|\bar{\boldsymbol{\mu}}_{\text{ctx}}\|_2 \in [4.73, 5.18]$ vs. $\sqrt{32} \approx 5.66$) at 0.06–0.16 anomaly divergence, whereas tap fusion and barycenter subtraction restore divergence to 0.70–0.94 and cut mean coordinate D_{KS} from 0.140 to 0.096.

VIII. CONCLUSION AND FUTURE WORK

We evaluated *Laminar-JEPA* and its autopsy-distilled counterpart *Laminar-JEPA-Lite* across 33 development and 118 untouched held-out telemetry streams (stream-level means unless noted):

1) *Flagship vs. distilled circuit-breaker panels:* On the 33-stream development benchmark, *Laminar-JEPA* and *Laminar-JEPA-Lite* reach buffered Point-F1 of 0.244 and 0.242 and causal ER of 67.3% and 65.0% at 1.3% and 1.0% FPR_{hold}, with *Laminar-JEPA-Lite* cutting parameters by 35%–52%, accelerating inference by 2.43×, and leading 7-dataset macro Point-F1 (0.171). On the 19-stream residual cache, the 224-dial panel lifts causal Point-F1 over 1D collapse (0.013 → 0.071, $p = 0.0023$), whereas the 160-dial subset (0.098) and training-whitened ℓ_2 split-conformal rule (0.072) match or exceed it at lower complexity. Across 118 untouched held-out streams under strictly disjoint checkpoint selection, *Laminar-JEPA-Lite* lifts causal Point-F1 from 0.074 to 0.085 ($p = 0.0034$), raises causal ER from 49.5% to 67.0% across 8/9 datasets (sign-flip $p = 0.012$), and remains positive in all 12/12 LODO folds on the 141-stream archive, while nominal quenching lowers four-horizon causal FPR_{hold} across 7/9 datasets (5.57% → 5.37%, median 0.96%, $p = 0.016$).

2) *Mechanisms and horizon latency:* Matched-backbone controls measure LayerNorm tail compression and barycenter drift reduction, while scalar R_{LT} modulation is null. On five exploratory streams ($S \in \{1, 4, 16, 32, 64\}$), both panels retain a 2.1×–4.3× stream-mean causal Point-F1 gain over 1D scoring down to zero reporting lag ($S = 1$), without establishing cross-domain latency rules.

3) *Calibration, single-sensor, and slow-ramp boundaries:* Stream-specific Sokal autocorrelation (τ_{int} up to 206) leaves expected tail counts of 1.5 ($q_{0.995}$) and 0.3 ($q_{0.999}$) at $N_{\text{eff}} = 300$; IQR and Beta-smoothed calibrators remain exploratory short-history remedies. Across 19 streams, an optional channel branch lifts injected single-sensor recall to 80.7%–96.5% while raising unmodified nominal causal FPR_{test} from 1.5% to 6.8%–19.4%. Controlled ramp sweeps confirm excess causal detection vanishes once ramp duration reaches $L \geq 1024$ ($4C$), explaining GHL's 0.000 Point-F1; pairing latent panels with raw-level trend monitors and testing natural sensor failures remain priorities for future work.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[11] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[12] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[13] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[14] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[15] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[16] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[17] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[18] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[19] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[20] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[21] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[22] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[23] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[24] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[25] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[26] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[27] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[28] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[29] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[30] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[31] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[32] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[33] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[34] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.

[35] H. Hotelling, "Multivariate quality control, illustrated by the air testing of sample bombsights," in *Techniques of Statistical Analysis*. McGraw-Hill, 1947, pp. 111–184.

[36] C. A. Lowry, W. H. Woodall, C. W. Champ, and S. E. Rigdon, "A multivariate exponentially weighted moving average control chart," *Technometrics*, vol. 34, no. 1, pp. 46–53, 1992.

[37] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[38] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[39] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.

[40] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.

[41] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[42] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[43] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[44] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[45] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[46] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[47] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[48] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[49] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.

[50] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.

[51] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.

[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.